

MariaDB

MariaDB is a forked version of MySQL. It was primarily developed due to concerns that arose when MySQL was acquired by Oracle Inc. MariaDB is a general-purpose DBMS engineered with extensible architecture to support a broad set of use cases via pluggable storage engines. It uses different storage engines to support different use cases.

MariaDB is an open source, multi-threaded, relational database management system, released under the GNU Public License (GPL). Its lead developer is Michael Monty Widenius, who is also one of the founders of MySQL AB. Many features contribute to MariaDB's popularity as a database system. Its speed is one of its most prominent features. MariaDB is also remarkably scalable, and is able to handle tens of thousands of tables and billions of rows of data. It can also manage small amounts of data quickly and smoothly, making it convenient for small businesses or personal projects. Another feature that sets it apart from its predecessors is a focus on security. MariaDB's built-in functions include those for manipulating and formatting text, business and statistical calculations, recording chronological information, as well as speciality features such as those related to GPS mapping.

MariaDB Server is one of the most popular open source databases in the world. It is available in Debian and Ubuntu, and is now the default database on Arch Linux, Manjaro, openSUSE, Red Hat Enterprise Linux, CentOS, Fedora and SUSE Linux Enterprise. And as one of the most broadly adopted and widely deployed in the world, MariaDB Server receives contributions from companies like Alibaba, Facebook and Google. Recently, Microsoft has also joined hands to support the MariaDB community.

Features of MariaDB

- MariaDB is available for use under GPL, LGPL and BSD.
- It includes a wide selection of storage engines, including high-performance storage engines, for working with other RDBMS data sources.
- It uses a standard and popular querying language.
- MariaDB runs on a number of operating systems and supports a wide variety of programming languages.
- It offers support for PHP, one of the most popular Web development languages.
- It offers Galera cluster technology.
- MariaDB also offers many operations and commands unavailable in MySQL, and eliminates/replaces features that impact performance negatively.

Other features include multi-source replication, fusion IO optimisations, table discovery and online alter table.

A comparative analysis of MariaDB and MySQL

MariaDB has a significantly high number of new features, which makes it better in terms of performance and user-

orientation. Let's evaluate both MariaDB and MySQL to determine which one is the better. The choice will ultimately depend on the IT managers and their familiarity with open source databases.

Some significant differences between MySQL and MariaDB

1. **Usage of the database:** Since 1995, MySQL has been regarded as the most implemented and widely used open source database till date. Many IT giants like Twitter, YouTube, Netflix and PayPal, as well as NASA, the US defence forces and Walmart, make use of this database. MariaDB, being a more recent arrival, is also taking strong root as back-end software in various IT giant organisations, such as Google, Red Hat, CentOS and Fedora.
2. **Structure of the database and the index:** MySQL is a pure relational database integrated with an ANSI-standard information schema, and consists of tables, columns, views, procedures, triggers, cursors, etc. The SQL of MySQL is a subset of ANSI SQL 99. MariaDB, on the other hand, is a fork of MySQL and, hence, has the same database structure and index. This facility makes MariaDB a strong choice for users who want to switch or upgrade their back-end directly without any issues of upgrading the database and data structures. Everything—from the data, table definitions, structures and APIs—remains identical when upgrading from MySQL to MariaDB.
3. **Binaries and implementation:** MySQL was developed using C and C++ and is fully compatible to run with almost every operating system like Microsoft Windows, MAC OS X, Linux, FreeBSD, UNIX, NetBSD, Novell Netware and many others. MariaDB was developed using C, C++, Bash and Perl. It is compatible with various operating systems like Microsoft Windows, Linux, MAC OS X, FreeBSD, Solaris, etc.
4. **Replication and clustering:** MySQL provides strong replication and clustering through master-master and master-slave replication, and makes use of the Galera Cluster for multi-master clustering. MariaDB provides almost the same replication and clustering facilities to end users as MySQL in terms of master-master and master-slave replication. It also uses Galera Cluster from version 10.1 onwards.
5. **Support for databases:** MySQL technical support services are provided round-the-clock via Oracle, and the support team consists of professional developers and engineers who provide various facilities like bug fixes, patches and version releases. Oracle offers MySQL premier support, extended support and sustaining support, depending on what users need. MariaDB provides strong support to users through